Day 12 – OSI Model

1. OSI MODEL

The Open Systems Interconnection (OSI) model is a conceptual framework developed by the International Organization for Standardization (ISO) to explain how data is communicated between devices over a network.

The OSI model divides network communication into seven layers. Each layer performs a specific function and works with the layers above and below it to enable successful data transmission.

It helps networking professionals understand communication processes, troubleshoot network problems, and identify potential security vulnerabilities.

2. Objectives

- Understand the seven layers of the OSI model.
- Learn the functions performed by each layer.
- Identify common protocols and networking devices associated with each layer.
- Understand how data travels from one device to another.
- Explore the importance of the OSI model in network security.

3. Seven Layers of the OSI Model

Layer 7 – Application Layer

Function: Provides network services to applications used by end users.

- Enables applications to communicate over a network.
- Supports services such as web browsing, email, and file transfers.
- Acts as the interface between network services and application software.

Protocols: HTTP, HTTPS, FTP, SMTP, DNS.

Example: A user opens a website using a web browser.

Cybersecurity relevance: Secure communication using HTTPS, protection against application-layer attacks, and identifying suspicious application traffic.

Layer 6 – Presentation Layer

Function: Ensures that data is represented in a format the receiving system can understand.

- Converts data between different formats.
- Supports data encryption and decryption.
- Performs data compression and decompression.

Examples: Character encoding, JPEG, PNG, and data-format conversion.

Cybersecurity relevance: Encryption and decryption help protect data confidentiality. Actual encryption can be implemented at different layers, depending on the protocol.

Layer 5 – Session Layer

Function: Establishes, manages, and terminates communication sessions between applications.

- Establishes communication sessions.
- Maintains and coordinates ongoing exchanges.
- Helps resume or terminate sessions when required.

Examples: Session management in application communication and RPC-based systems.

Cybersecurity relevance: Session management helps protect against session hijacking and unauthorized session reuse.

Layer 4 – Transport Layer

Function: Provides end-to-end communication between applications running on different devices.

- Divides data into smaller units for transmission.
- Uses port numbers to identify applications.
- Supports reliability, flow control, and error recovery where provided by the protocol.

Protocols: TCP and UDP.

Example: TCP supports reliable delivery of data, while UDP provides connectionless transport with lower overhead.

Cybersecurity relevance: Firewalls can filter traffic using port numbers and transport protocols. Attackers may also exploit exposed ports and transport-layer weaknesses.

Layer 3 – Network Layer

Function: Transfers packets between different networks using logical addressing and routing.

- Uses IP addresses to identify network destinations.
- Determines routes for packet delivery.
- Supports forwarding packets through routers.

Protocols: IPv4, IPv6, and ICMP.

Device: Router.

Cybersecurity relevance: IP filtering, network segmentation, and monitoring suspicious traffic help protect this layer.

Layer 2 – Data Link Layer

Function: Transfers frames between devices on the same local network.

- Uses MAC addresses for local delivery.
- Organizes data into frames.
- Detects certain transmission errors.
- Controls access to the network medium where applicable.

Technologies and protocols: Ethernet and Wi-Fi MAC functions.

Devices: Switches and network bridges.

Cybersecurity relevance: MAC spoofing, ARP-related attacks, and unauthorized access to local networks are important security concerns.

Layer 1 – Physical Layer

Function: Transmits raw bits through a physical or radio communication medium.

- Defines electrical, optical, radio, and physical signaling characteristics.
- Specifies cables, connectors, and transmission rates.
- Carries bits between connected devices.

Examples: Ethernet cables, fiber-optic cables, radio signals, and repeaters.

Cybersecurity relevance: Physical security helps prevent cable tampering, unauthorized device connections, and physical access to networking equipment.

4. OSI Model Chart

Layer	Layer Name	Data Unit	Common Examples
7	Application	Data	HTTP, HTTPS, DNS, SMTP
6	Presentation	Data	JPEG, PNG, data encoding
5	Session	Data	Session management, RPC
4	Transport	Segment / Datagram	TCP, UDP
3	Network	Packet	IPv4, IPv6, ICMP
2	Data Link	Frame	Ethernet, Wi-Fi MAC
1	Physical	Bits	Cables, fiber, radio signals

Note: These data-unit names are commonly used in the OSI model. Actual protocol stacks do not always map perfectly to the seven layers.

5. How Data Travels Through the OSI Model

When a user sends information over a network, the data typically passes through the layers in the following order:

1. **Application:** The application creates the information to be sent.

2. **Presentation:** Data may be formatted, encoded, compressed, or encrypted.
3. **Session:** Communication sessions are managed where applicable.
4. **Transport:** Data is divided into transport units, and ports identify the relevant applications.
5. **Network:** Packets receive logical addressing and are routed toward the destination.
6. **Data Link:** Frames are prepared for transmission across the local link.
7. **Physical:** Bits are transmitted through the communication medium.

At the receiving device, the data is processed in the reverse direction, from Layer 1 to Layer 7. This process is called **decapsulation** at the receiving end, while adding layer-specific information during transmission is called **encapsulation**.

6. Importance of the OSI Model in Cybersecurity

- Helps locate the layer where a network problem occurs.
- Makes it easier to understand network protocols and communication.
- Helps security teams classify attacks according to their effects.
- Supports firewall configuration and network monitoring.
- Improves understanding of secure communication and encryption.
- Helps identify risks involving physical devices, local networks, IP traffic, transport ports, and applications.

The OSI model is an important foundation of computer networking and cybersecurity. Its seven layers explain how data is transmitted, processed, and received across networks. Understanding each layer helps students and security professionals troubleshoot communication issues, understand protocols, and identify possible security risks.